

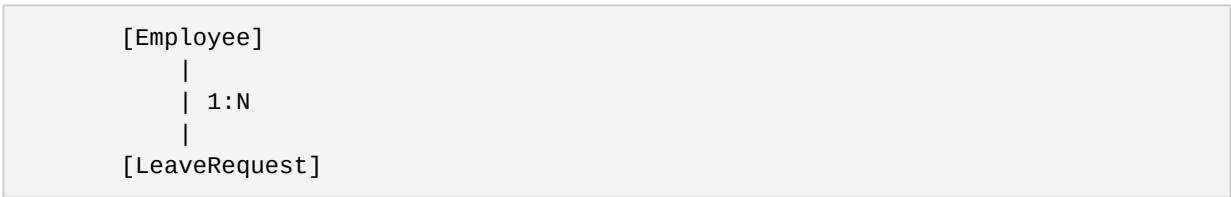
Data Model Documentation

Overview

The system manages the following core domains:

- Employee: Responsible for storing employee details and managing leave requests.
- LeaveRequest: Responsible for tracking leave requests submitted by employees.

Entity Relationship Diagram



Entities

Employee

Purpose: Represents an employee within the organization, managing their personal details and leave balance.

Table/Collection Name: Employee

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
name	VARCHAR	Yes	Max 255 chars	Employee's full name
email	VARCHAR	Yes	Max 255 chars, unique	Employee's email address
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many LeaveRequest (1:N): Inverse via LeaveRequest . employee_id

Validation Rules:

- name must be non-empty
- email must be non-empty and unique across all records

LeaveRequest

Purpose: Represents a leave request submitted by an employee, tracking its status and dates.

Table/Collection Name: LeaveRequest

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
employee_id	UUID	Yes	Foreign key to Employee . id	Employee who submitted the request
start_date	DATE	Yes	Must not be in the past	Start date of the leave
end_date	DATE	Yes	Must be after start_date	End date of the leave
status	VARCHAR	Yes	Must be one of "PENDING", "APPROVED", "REJECTED"	Current status of the leave request
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references Employee . id

Validation Rules:

- start_date must be a valid date and not in the past
- end_date must be a valid date and after start_date
- status must be one of "PENDING", "APPROVED", "REJECTED"

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
LeaveRequest	employee_id	Employee	id	N:1	CASCADE

Unique Constraints

- `Employee(email)`: Ensures no duplicate email addresses

Data Validation Rules

Business Rules:

- `LeaveRequest.start_date`: Must not be in the past, otherwise return a validation error.
- `LeaveRequest.end_date`: Must be after `start_date`, otherwise return a validation error.
- `LeaveRequest.status`: Must be one of "PENDING", "APPROVED", "REJECTED", otherwise return a validation error.

Format Rules:

- `Employee.email`: Must match the email format regex
`^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`

Data Lifecycle

Creation

- A new `Employee` is created when a new user registers.
- Default status for `LeaveRequest` is `PENDING`.
- `created_by` is automatically set to the current user

Updates

- `Employee.email` can only be updated by the employee themselves.
- `updated_at` is automatically updated on any modification to an `Employee` or `LeaveRequest`.

Archival / Soft Delete

- Records are not archived but can be marked as deleted by setting the status to `DELETED`.

Indexes

- Primary key: id
- Foreign keys: employee_id
- Search fields: name, email

Example Records

Employee

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "John Doe",
  "email": "john.doe@example.com",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```

LeaveRequest

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "start_date": "2026-01-15",
  "end_date": "2026-01-20",
  "status": "PENDING",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```